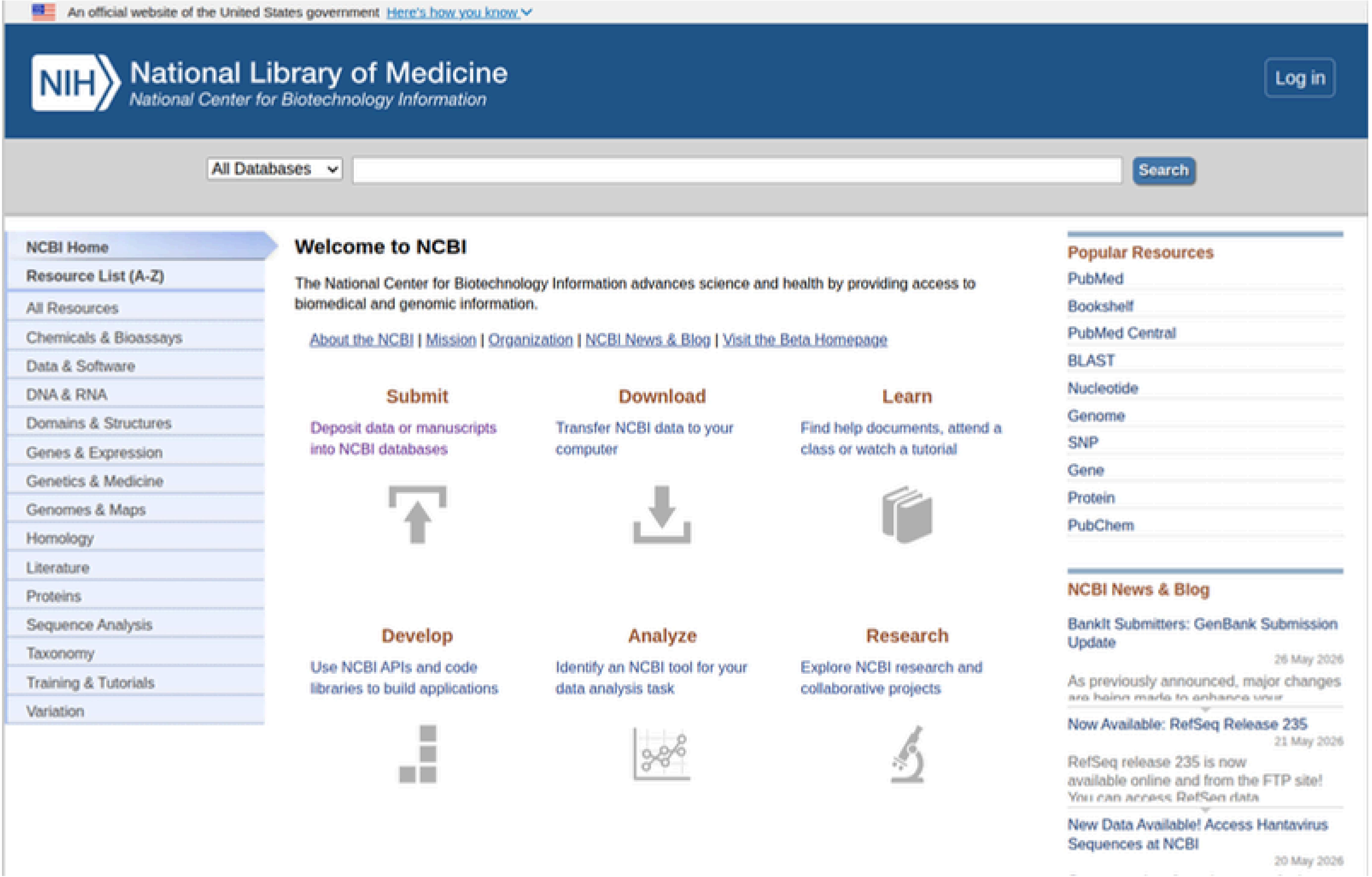
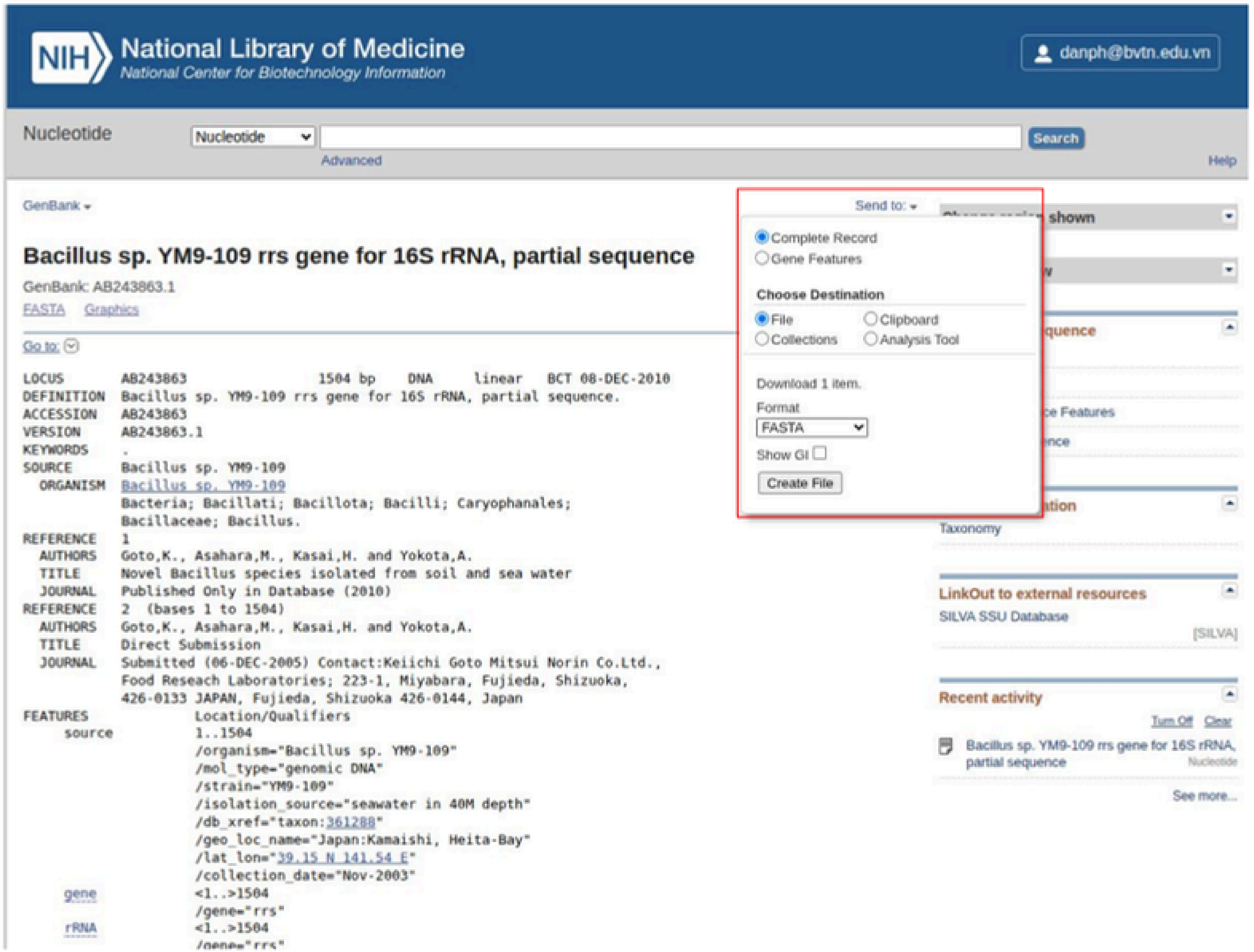
Lecture 06

Programmatic Database Access with EDirect

What we previously learned



What we previously learned



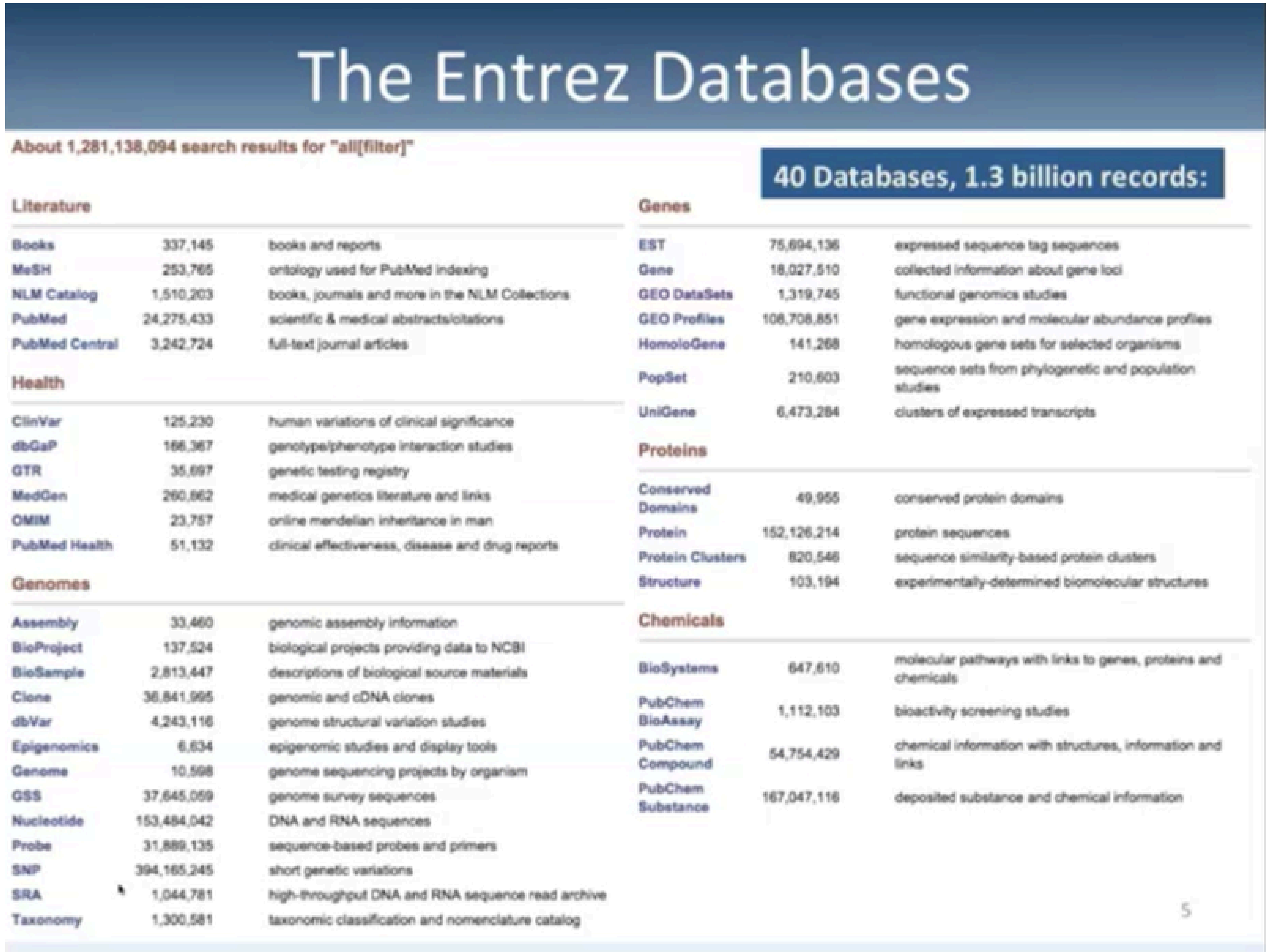
Download FASTA

What if we want to download 100 files at a time??

The old method will consume lots of time

=> Use NCBI EDirect

NCBI Entrez Direct - NCBI EDirect



The Entrez Databases

About 1,281,138,094 search results for "all[filter]"

40 Databases, 1.3 billion records:

Literature

Books	337,145	books and reports
MeSH	253,765	ontology used for PubMed indexing
NLM Catalog	1,510,203	books, journals and more in the NLM Collections
PubMed	24,275,433	scientific & medical abstracts/citations
PubMed Central	3,242,724	full-text journal articles

Health

ClinVar	125,230	human variations of clinical significance
dbGaP	166,367	genotype/phenotype interaction studies
GTR	35,697	genetic testing registry
MedGen	260,862	medical genetics literature and links
OMIM	23,757	online mendelian inheritance in man
PubMed Health	51,132	clinical effectiveness, disease and drug reports

Genomes

Assembly	33,460	genomic assembly information
BioProject	137,524	biological projects providing data to NCBI
BioSample	2,813,447	descriptions of biological source materials
Clone	36,841,995	genomic and cDNA clones
dbVar	4,243,116	genome structural variation studies
Epigenomics	6,634	epigenomic studies and display tools
Genome	10,598	genome sequencing projects by organism
GSS	37,645,059	genome survey sequences
Nucleotide	153,484,042	DNA and RNA sequences
Probe	31,889,135	sequence-based probes and primers
SNP	394,165,245	short genetic variations
SRA	1,044,781	high-throughput DNA and RNA sequence read archive
Taxonomy	1,300,581	taxonomic classification and nomenclature catalog

Genes

EST	75,694,136	expressed sequence tag sequences
Gene	18,027,510	collected information about gene loci
GEO DataSets	1,319,745	functional genomics studies
GEO Profiles	108,708,851	gene expression and molecular abundance profiles
HomoloGene	141,268	homologous gene sets for selected organisms
PopSet	210,603	sequence sets from phylogenetic and population studies
UniGene	6,473,284	clusters of expressed transcripts

Proteins

Conserved Domains	49,955	conserved protein domains
Protein	152,126,214	protein sequences
Protein Clusters	820,546	sequence similarity-based protein clusters
Structure	103,194	experimentally-determined biomolecular structures

Chemicals

BioSystems	647,610	molecular pathways with links to genes, proteins and chemicals
PubChem BioAssay	1,112,103	bioactivity screening studies
PubChem Compound	54,754,429	chemical information with structures, information and links
PubChem Substance	167,047,116	deposited substance and chemical information

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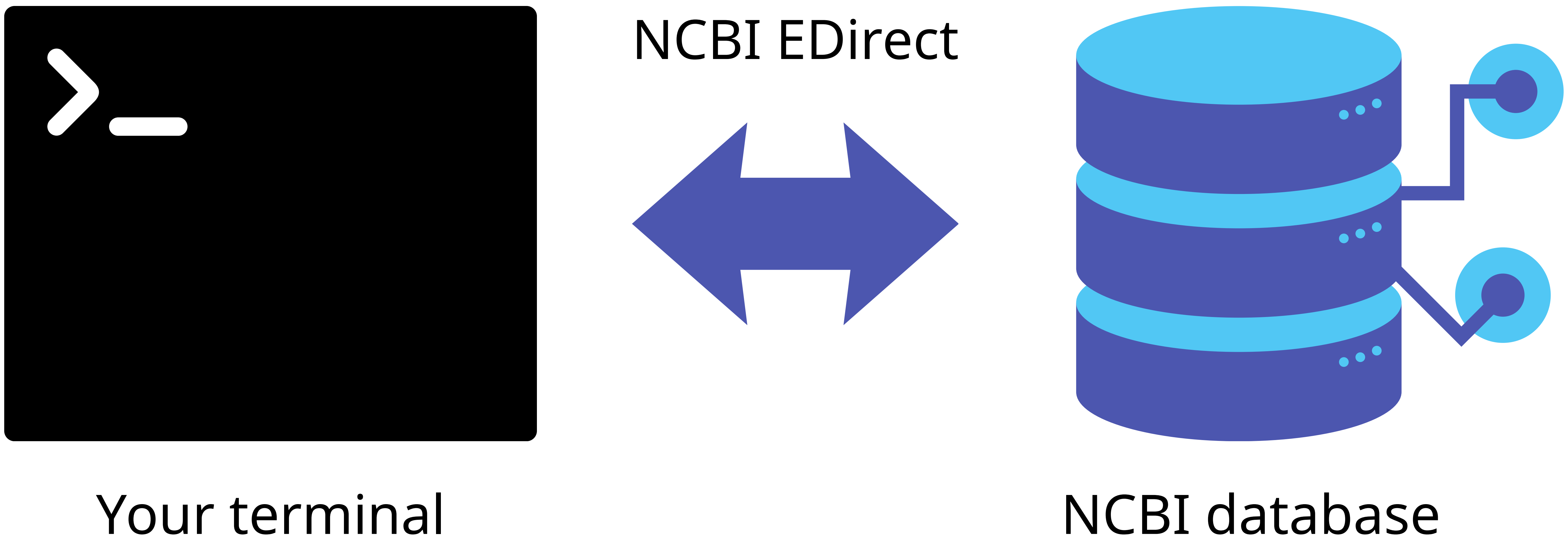
- Entrez is a molecular biology database system that provides integrated access to nucleotide and protein sequence data, gene-centered and genomic mapping information, 3D structure data, PubMed MEDLINE, and more

NCBI Entrez Direct - NCBI EDirect

NCBI Entrez Direct (EDirect) is a suite of command-line tools that allows you to interact with NCBI's massive biological databases directly from your terminal



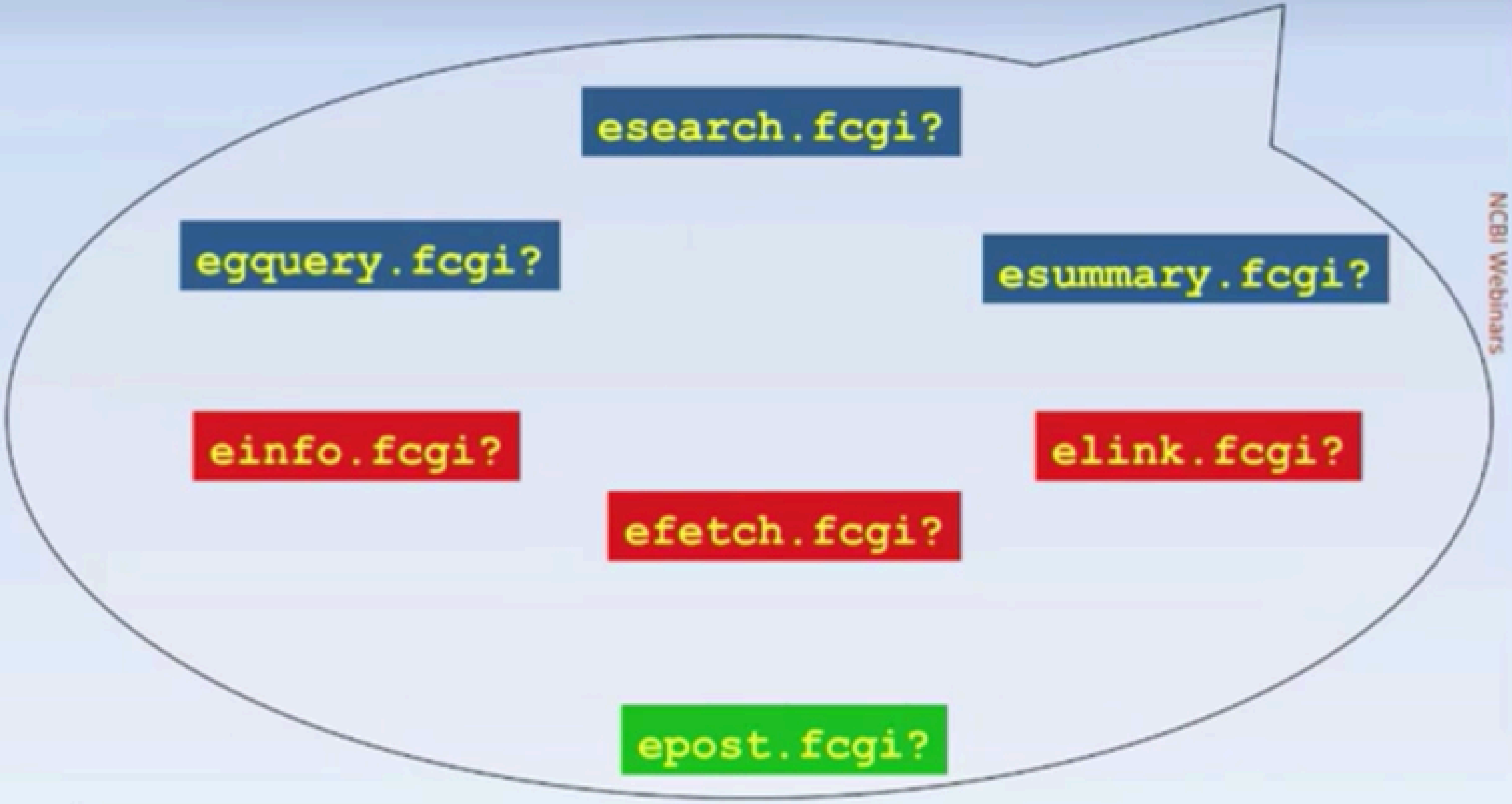
What is Entrez Database

Entrez Databases and UIDs

Entrez Database	UID common name	E-utility Database Name
BioProject	BioProject ID	bioproject
BioSample	BioSample ID	biosample
Biosystems	BSID	biosystems
Books	Book ID	books
Conserved Domains	PSSM-ID	cdd
dbGaP	dbGaP ID	gap
dbVar	dbVar ID	dbvar
Epigenomics	Epigenomics ID	epigenomics
EST	GI number	nucest
Gene	Gene ID	gene
Genome	Genome ID	genome
GEO Datasets	GDS ID	gds
GEO Profiles	GEO ID	geoprofiles
GSS	GI number	nucgss
HomoloGene	HomoloGene ID	homologene
MeSH	MeSH ID	mesh
NCBI C++ Toolkit	Toolkit ID	toolkit
NCBI Web Site	Web Site ID	ncbisearch
NLM Catalog	NLM Catalog ID	nlmcatalog
Nucleotide	GI number	nuccore
PopSet	PopSet ID	popset
Probe	Probe ID	probe
Protein	GI number	protein
Protein Clusters	Protein Cluster ID	proteinclusters
PubChem BioAssay	AID	pcassay
PubChem Compound	CID	pccompound
PubChem Substance	SID	pcsubstance
PubMed	PMID	pubmed
PubMed Central	PMCID	pmc
SNP	rs number	snp
SRA	SRA ID	sra
Structure	MMDB-ID	structure
Taxonomy	TaxID	taxonomy
UniGene	UniGene Cluster ID	unigene

The E-utilities

http://eutils.ncbi.nlm.nih.gov/entrez/eutils/ *eutil*.fcgi?



Requirement for using EDirect

Good internet

Web browser

Programming language

NCBI Entrez Direct - NCBI EDirect

- EDirect is not one single command, but a suite of modular "navigation programs" that communicate with each other
- EDirect has many commands follow here for details:

https://www.ncbi.nlm.nih.gov/books/NBK179288/

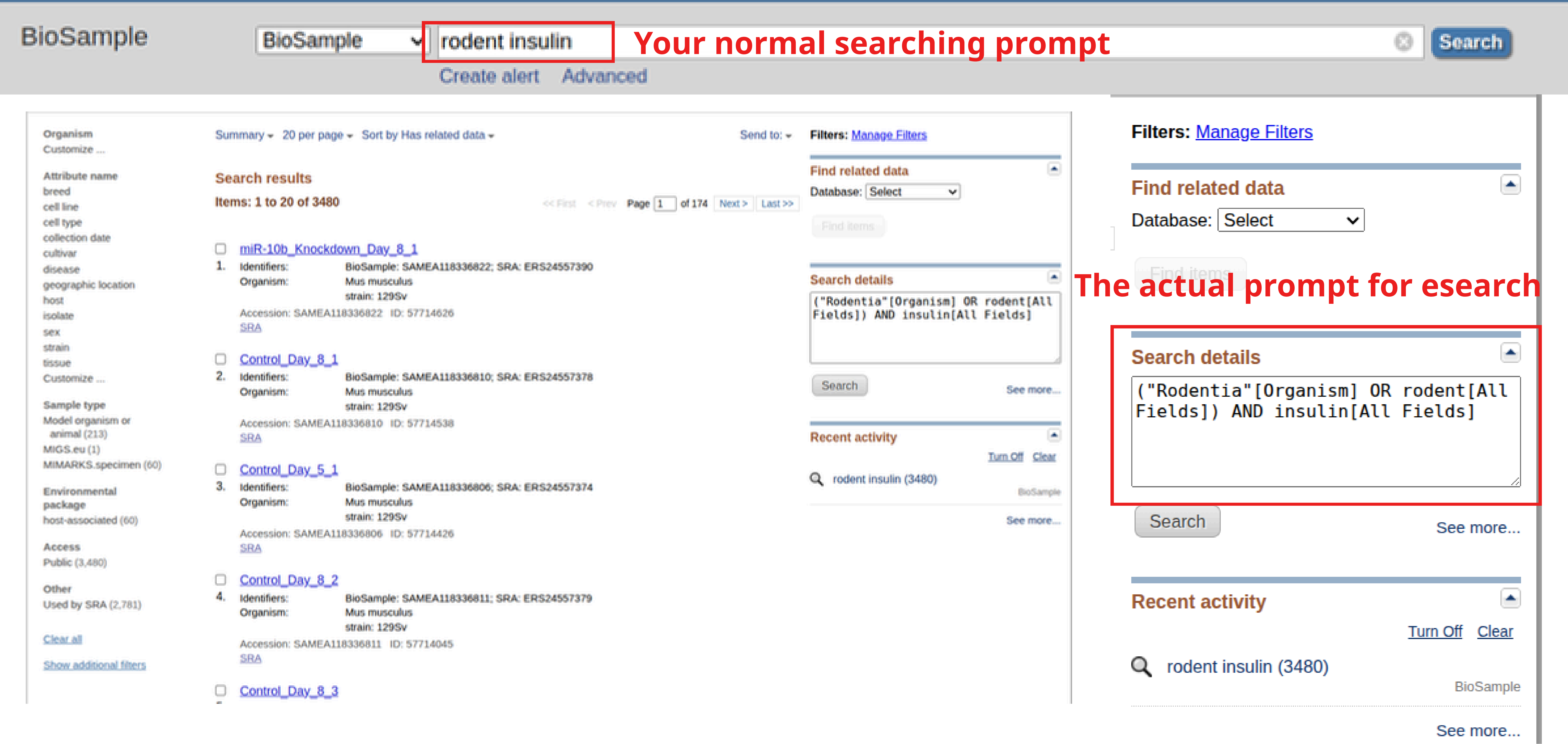
EDirect Utilities

text query	esearch	Set of UIDs
Set of UIDs	esummary	Record summary
Set of UIDs	efetch	Data record
Set of UIDs dbA	elink	Set of UIDs dbB
	einfo	
	xtract	

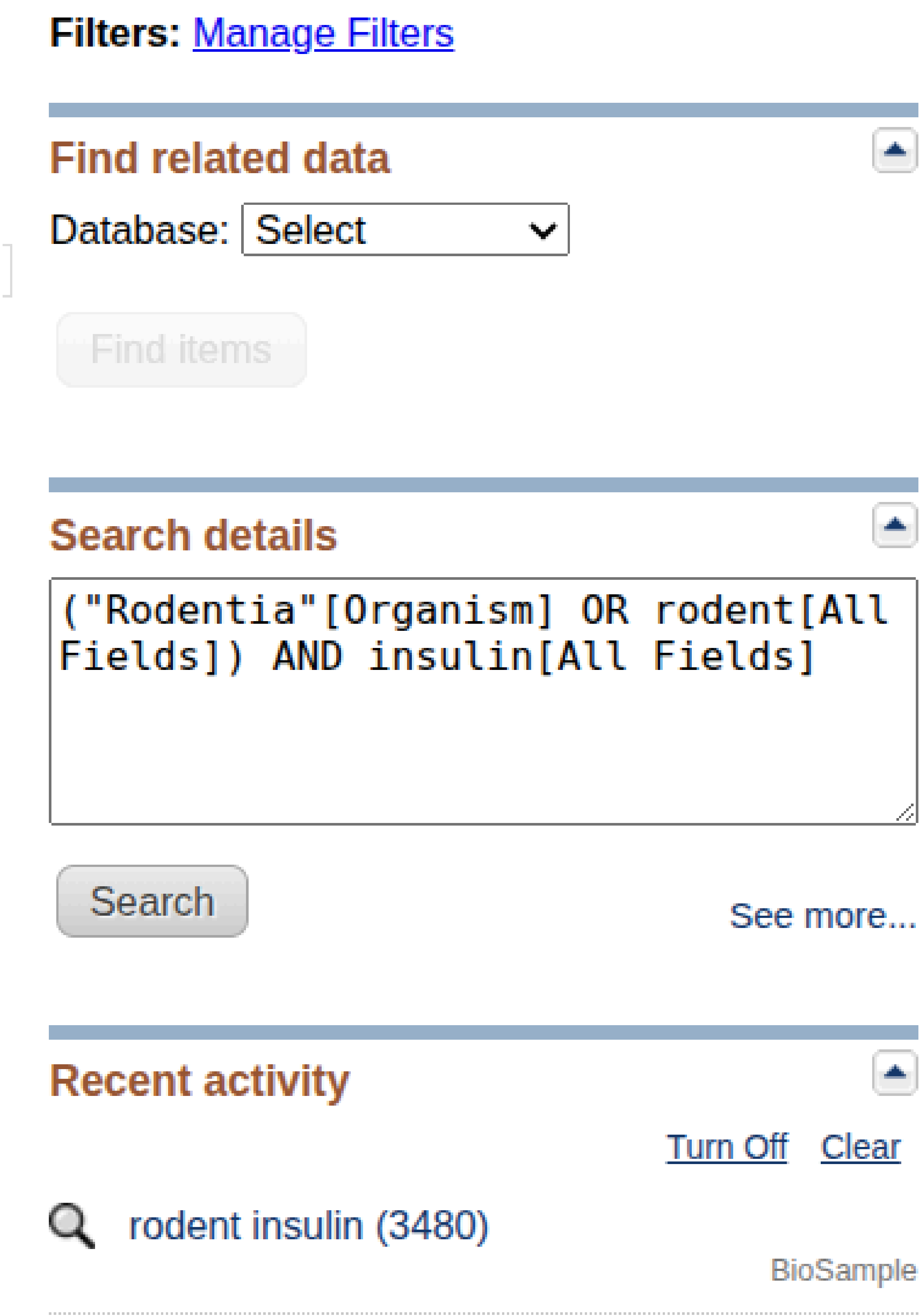
EDirect Utilities - Web equivalent

Web action	E-utility equivalent
PubMed search	**ESearch → ESummary**
Click on title in search results	EFetch
Click to full text	ELink
Gene search	**ESearch (→ ESummary/EFetch)**
Find all RefSeq proteins for the gene	ELink

Basic utilities: esearch



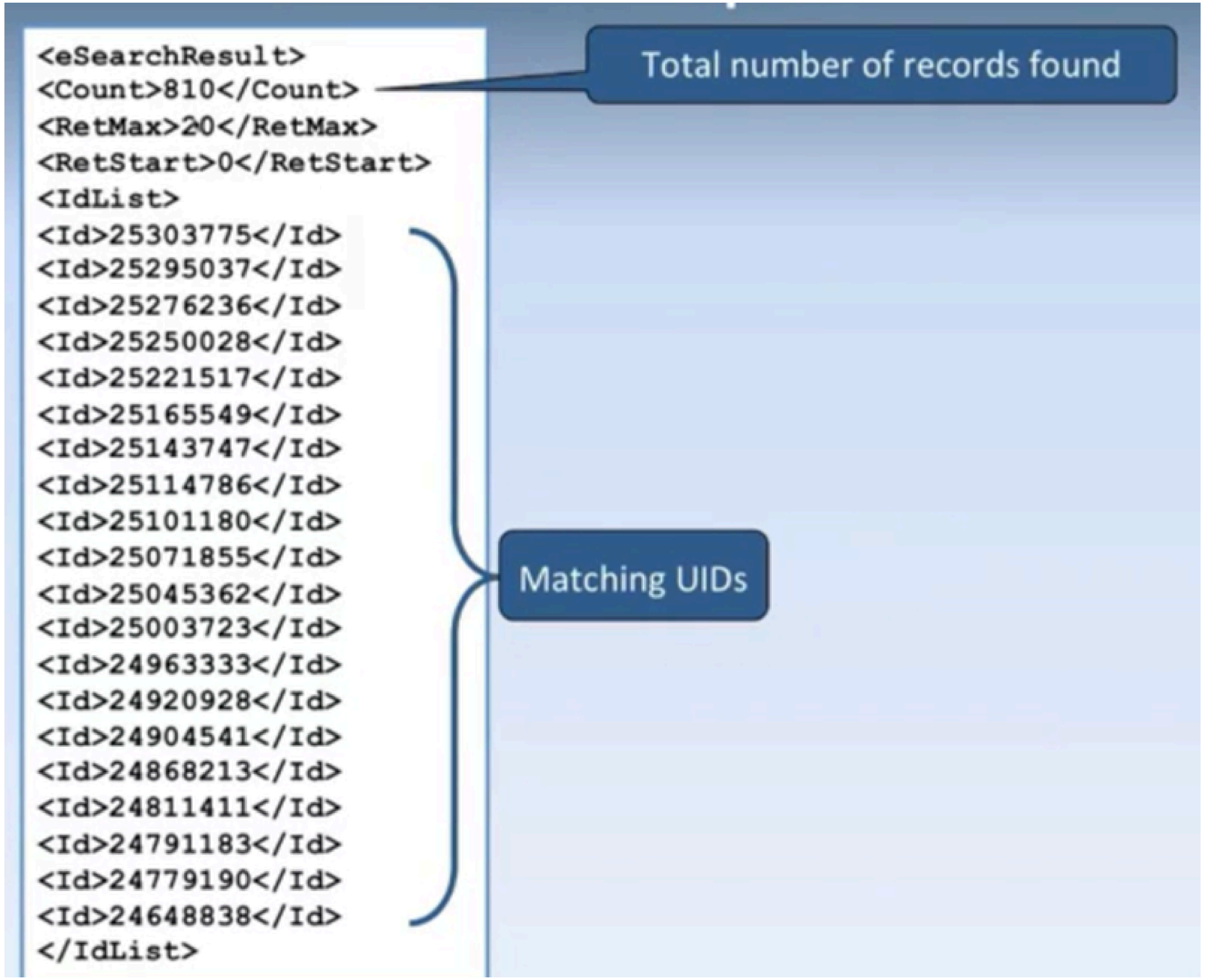
Basic utilities: esearch



If you search NCBI for just the word "rodent", you will get millions of results. It will return the gene, proteins, papers written by people named rodent, and sequences from random bacteria where the author just mentioned the word in the notes.

To fix this, we use **Field Tags**. By putting a tag in brackets next to our word, we force NCBI to only look in that specific category.

Ex: Salmonella[Organism] AND mitochondrion [Title]
=> Search for article that has “mitochondrion” in it for Salmonella spp.



Basic utilities: esearch

- esearch give a list of UIDs
- We can not know what are these samples, what to do with them

=> Use **esummary** to get there information

Basic utilities: esummary

```
(base) phuctran@schmoopy:~$ esearch -db nucleotide -query "insulin [GENE]"
<ENTREZ_DIRECT>
  <Db>nuccore</Db>
  <WebEnv>MCID_6a1dec37c1e0503bfc0f6481</WebEnv>
  <QueryKey>1</QueryKey>
  <Count>6</Count>
  <Step>1</Step>
  <Elapsed>1</Elapsed>
</ENTREZ_DIRECT>
```

No information or metadata

```
base) phuctran@schmoopy:~$ esearch -db nucleotide -query "insulin [GENE]" | esummary
```


```
?xml version="1.0" encoding="UTF-8" ?>
!DOCTYPE DocumentSummarySet>
DocumentSummarySet status="OK">
 <DocumentSummary>
   <Id>202471</Id>
   <Caption>M57671</Caption>
   <Title>Octodon degus insulin mRNA, complete cds</Title>
   <Extra>gi|202471|gb|M57671.1|OCOINS</Extra>
   <Gi>202471</Gi>
   <CreateDate>1991/03/01</CreateDate>
   <UpdateDate>1991/03/01</UpdateDate>
   <TaxId>10160</TaxId>
   <Slen>432</Slen>
   <Biomol>mRNA</Biomol>
   <MolType>rna</MolType>
   <Topology>linear</Topology>
   <SourceDb>insd</SourceDb>
   <SubType>tissue_type</SubType>
   <SubName>pancreas</SubName>
   <GeneticCode>1</GeneticCode>
   <Strand>ss</Strand>
   <Organism>Octodon degus</Organism>
   <Statistics>
     <Stat type="Length" count="432"/>
     <Stat type="Length" subtype="literal" count="432"/>
     <Stat type="all" count="5"/>
     <Stat type="blob_size" count="2763"/>
     <Stat type="cdregion" count="1"/>
     <Stat type="cdregion" subtype="CDS" count="1"/>
     <Stat type="gene" count="1"/>
     <Stat type="gene" subtype="Gene" count="1"/>
     <Stat type="imp" count="2"/>
     <Stat type="imp" subtype="polyA_signal" count="1"/>
     <Stat type="imp" subtype="polyA_site" count="1"/>
     <Stat type="org" count="1"/>
     <Stat type="pub" count="1"/>
     <Stat type="pub" subtype="PubMed" count="1"/>
     <Stat source="CDS" type="all" count="5"/>
```


Basic utilities: esummary

Still unreadable!!!

Pipe to **xtract** to only get the necessary field

Basic utilities: esummary

```
(base) phuctran@schmoopy:~$ esearch -db nucleotide -query "insulin [GENE]" | esummary | xtract -pattern DocumentSummary -element Caption,Slen,Title
M57671  432     Octodon degus insulin mRNA, complete cds
M26159  361     Rat insulin I 5' flanking region
NM_001204686    968     Aplysia californica insulin precursor (PIN), mRNA
NW_004798128    314614  Aplysia californica isolate F4 #8 unplaced genomic scaffold, AplCal3.0 scaffold00858, whole genome shotgun sequence
NTJE000000000   1213011 Eschrichtius robustus isolate AM-GW-M-EMB-4M-20130531, whole genome shotgun sequencing project
JWIN03000075    8536121 Camelus dromedarius breed African isolate Drom800 Contig74, whole genome shotgun sequence
```

Pipe to **xtract** to only get Caption, Slen, Title

Classwork 1: Use esearch and search for human COX1 gene from nucleotide database and summary it with only Caption, CreateDate, Title and view only the first 10 results

Basic utilities: efetch

- **esearch** only finds the ID numbers
- **efetch** retrieves the actual biological record
- **esearch** just gives us a count of how many things it found. To actually look at the DNA or read the paper, we need to hand those results over to efetch using the pipe symbol and use efetch

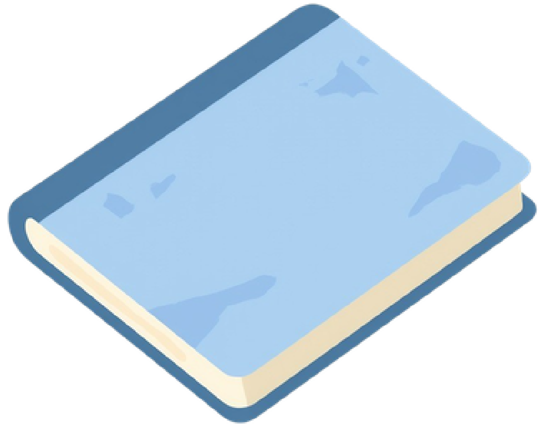
esearch will find the title of the books

efetch will get those books base on the title

Basic utilities: efetch

- EDirect is a combination of databases, so it has many type of file format for 1 UIDs. We need to use ***-format*** to get the desired format

esearch will find the title of the books



FASTA

efetch -format fasta

GB

efetch -format gb

Download multiple files with EDirect

- **esearch** find the IDs for your query
- **efetch** download all those IDs from **esearch** results without have to make a loop command

Classwork 2: Use efetch and download the first 20 human COX1 gene FASTA sequence from nucleotide database and export to a filename “human_cox1.fasta”

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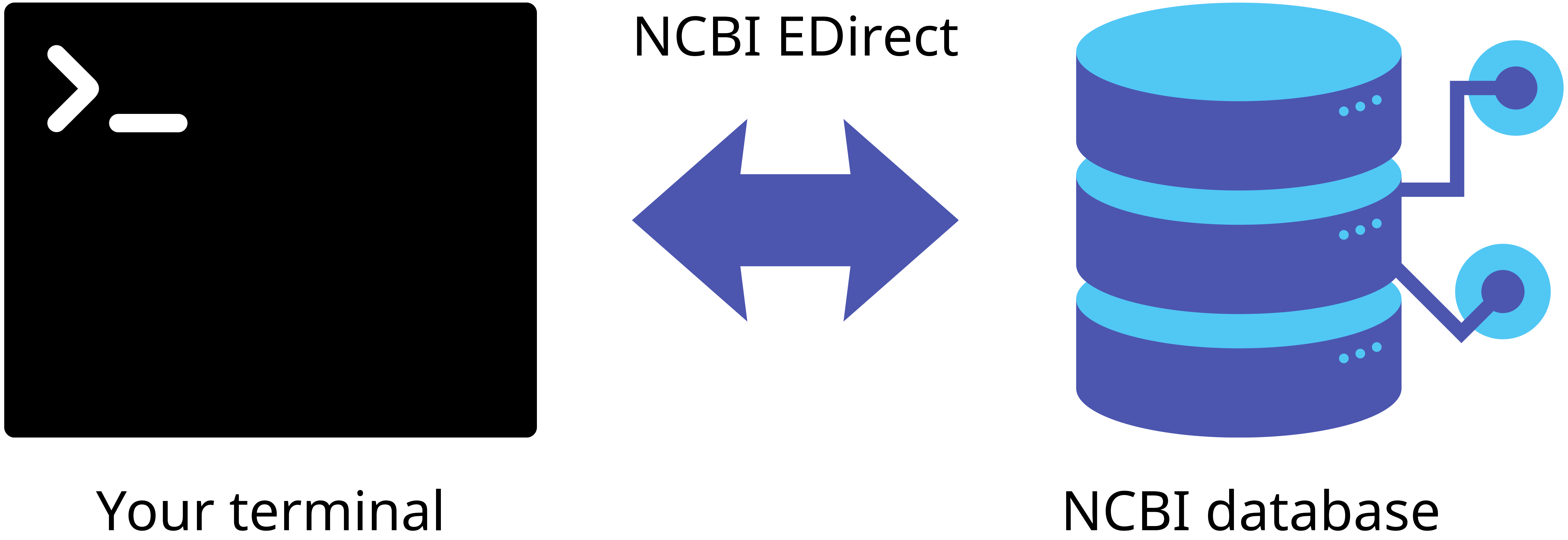
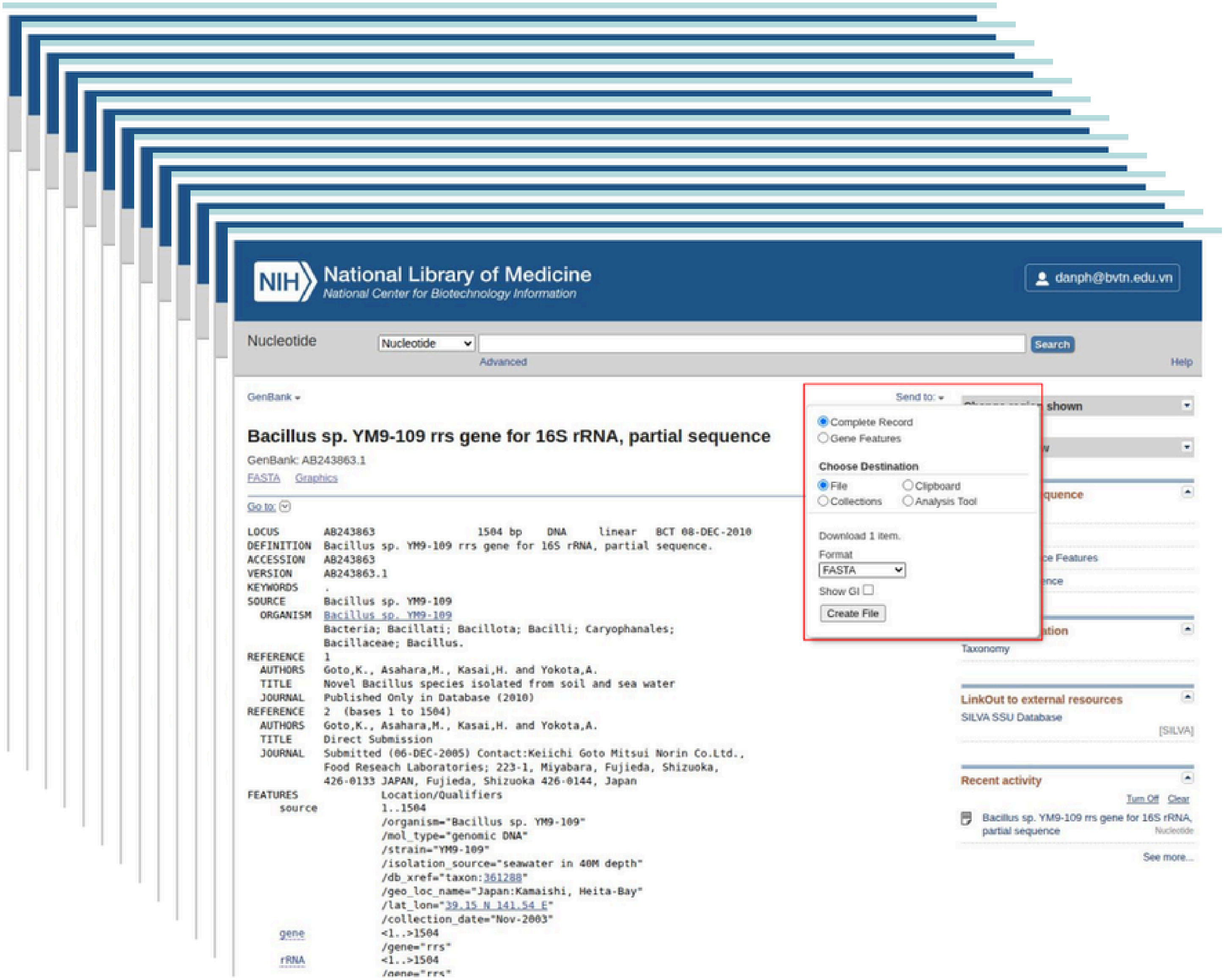
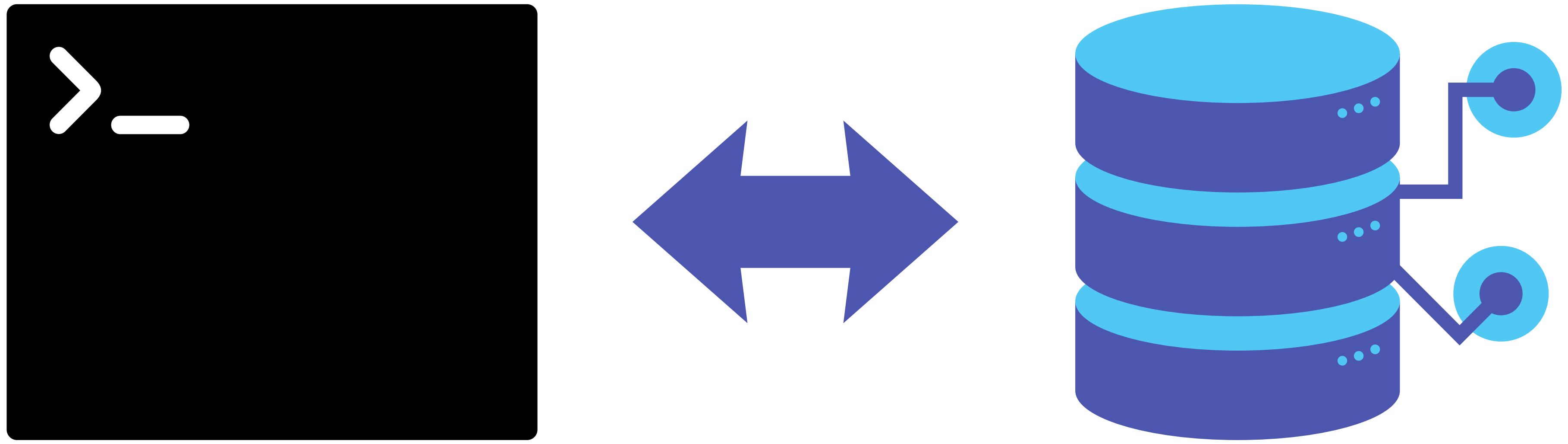


Why using NCBI EDirect?





Clicking multiple times

Some command lines for multiple files